Global Layoffs Storytelling Dashboard

Introduction

This project is an interactive data storytelling dashboard developed using Python, Streamlit, Pandas, and Plotly. The purpose of the project is to analyze global layoffs data and present meaningful insights through visualizations and storytelling techniques.

The dashboard helps users understand how layoffs affected different industries, companies, and countries over time. Instead of displaying raw data in tables, the project converts information into interactive graphs and business insights.

Objective

The main objectives of this project are:

- To analyze global layoffs data
- To identify trends and patterns
- To create interactive visualizations
- To present insights using storytelling techniques
- To improve understanding of business and economic impacts

Technologies Used

- Python
- Streamlit
- Pandas
- Plotly
- NumPy

Dataset Used

The project uses a global layoffs dataset containing information such as:

- Company name
- Industry
- Country
- Total employees laid off
- Date of layoffs
- Funding details

The dataset was cleaned and processed before visualization.

Features of the Dashboard

The dashboard contains:

- Interactive filters
- KPI cards
- Bar charts
- Pie charts
- Line graphs
- Scatter plots
- Storytelling sections

Users can filter data by industry and country to explore different insights dynamically.

Key Insights

- Technology companies experienced the highest layoffs.
- The United States had the largest number of layoffs globally.
- Layoffs increased significantly during economic slowdown periods.
- Even highly funded companies conducted layoffs.
- Visualization made complex business data easier to understand.

Learning Outcomes

Through this project, I learned:

- Data cleaning and preprocessing
- Interactive dashboard development

- Storytelling using visual analytics
- Data visualization techniques
- Business insight generation using Python

Conclusion

This project demonstrated how data storytelling and visualization can transform raw datasets into meaningful business insights. Interactive dashboards improve understanding, communication, and decision-making by presenting data in a clear and engaging format.

Overall, the project helped improve my practical knowledge of Python-based data analysis and visualization tools.